

AUDSPECT

Breach & Attack Simulation Platform

Adversary Simulation Assessment Report

Atomic Red Team - Selective Techniques

RISK: CRITICAL

Prepared for	WIN-MHQITQ23L3Q
Endpoint	Windows/amd64 (10.0.17763)
Environment	Production
Assessment date	11 June 2026, 06:40 UTC
Report generated	11 June 2026, 06:59 UTC
Run reference	18b7f39d

1. Executive Summary

PRODUCTION ENVIRONMENT

This assessment executed live adversary techniques against a production endpoint. Endpoint changes were reverted where the snapshot system captured them — see section 7 (Changes & Cleanup Verification).

This report presents the results of an automated breach & attack simulation executed against WIN-MHQITQ23L3Q (Windows/amd64 (10.0.17763)) on 11 June 2026. The assessment ran 298 MITRE ATT&CK-aligned techniques to measure how effectively the endpoint's security controls prevent real adversary behaviour.

Of the techniques executed, 50 were prevented or blocked by a security control and 111 executed successfully without being stopped. This yields a prevention effectiveness of 32% and an overall risk classification of Critical (risk score 100/100). Consecutive failures across adjacent kill-chain phases amplified exposure by 1.8×, indicating an attacker could chain multiple steps with limited resistance. A further 94 technique(s) could not be evaluated (execution errors — malformed test content, timeouts, missing prerequisites or scheduler contention) and 43 were skipped; both are excluded from the prevention and exposure scores so they neither inflate nor deflate the result.

100 critical or high-severity technique(s) were not prevented and require prioritised remediation (see Key Findings). Testing exercised 7 of the 14 MITRE ATT&CK Enterprise tactics across 298 technique(s) — this is the breadth of THIS run, not a measure of overall MITRE ATT&CK coverage. Broadening scenario coverage in future runs will increase assurance. Prioritised recommendations are provided in section 8.

Overall posture: Critical (risk score 100 / 100, Baseline)

298 techniques · 50 prevented · 111 succeeded · 94 errored · 43 skipped · 100 critical/high finding(s)

Prevention & Detection

111 technique(s) executed without being prevented. A FAIL means a prevention control did not stop execution — not that the attacker's end objective was independently verified. Of those that executed:

0

Detected

an alert fired (Microsoft Defender) — a SOC would see this

0

Logged only

telemetry exists but no alert was raised

111

Undetected

no telemetry observed — executed unseen (worst case)

Business-Objective Risk

What an attacker could achieve against this endpoint, by objective — derived from the tested techniques in each area (execution errors excluded).

Credential Theft	LOW RISK	0 of 9 technique(s) unprevented
Privilege Escalation	HIGH RISK	10 of 18 technique(s) unprevented
Persistence	HIGH RISK	9 of 9 technique(s) unprevented
Command & Control	HIGH RISK	8 of 13 technique(s) unprevented
Defense Evasion	HIGH RISK	69 of 72 technique(s) unprevented
Code Execution	MEDIUM RISK	4 of 17 technique(s) unprevented
Discovery	MEDIUM RISK	11 of 23 technique(s) unprevented

2. Assessment Scorecard



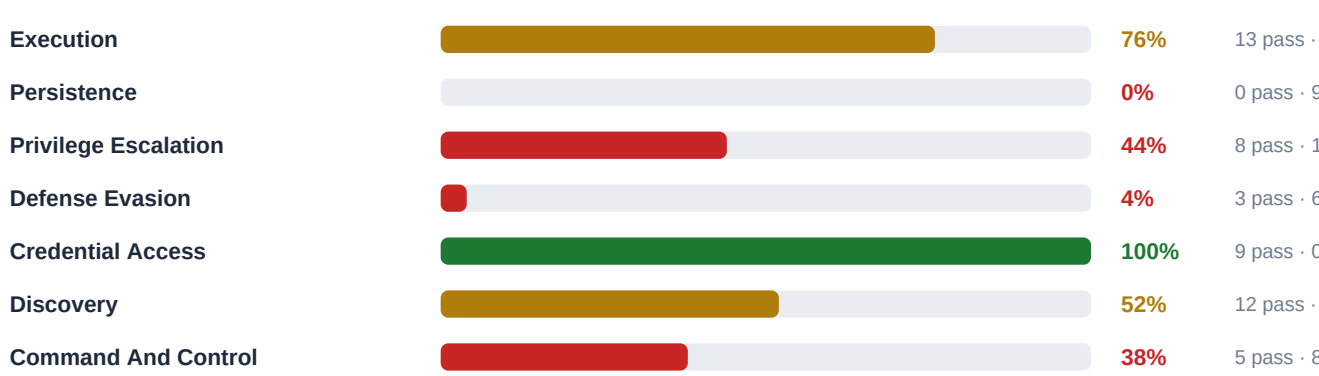
3. Methodology & Scope

Framework	MITRE ATT&CK (Enterprise)
Engines	Atomic Red Team
Target endpoint	WIN-MHQITQ23L3Q — Windows/amd64 (10.0.17763)
Environment	Production
Security context	Microsoft Defender (real-time protection ON)
Techniques executed	298
Tactics exercised	7 of 14 ATT&CK tactics
Window	11 Jun 2026 06:40 UTC

Security context lists the AV/EDR products detected on the endpoint (presence only). Detection by third-party EDR is not locally observable and is therefore never asserted in this report.

Each technique is scored as PASS when a security control prevented or blocked it, FAIL when it executed without being stopped, ERROR when the test itself could not execute correctly (malformed content, timeout, missing prerequisite, scheduler contention), or SKIPPED when it was not run. Only PASS and FAIL count toward the score: a FAIL is a finding — the simulated adversary behaviour succeeded against this endpoint — while ERROR and SKIPPED are excluded because they reflect a test-execution problem, not the endpoint's defences.

4. ATT&CK Tactic Breakdown



5. Key Findings

HIGH	F-01 T1059.001 — Command and Scripting Interpreter: PowerShell	
	Execution	
	What happened	Security control did not prevent this technique — it executed without being blocked.
	Remediation	1) Enable PowerShell Script Block Logging and Constrained Language Mode. 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution. 3) Enable ASR rules targeting script-based and macro-based execution. 4) Monitor for anomalous parent-child process relationships in EDR.

HIGH**F-02 T1059.003 — Command and Scripting Interpreter: Windows Command Shell** Defensive Techniques
Execution**What happened** Security control did not prevent this technique — it executed without being blocked.**Remediation**

- 1) Enable PowerShell Script Block Logging and Constrained Language Mode.
- 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution.
- 3) Enable ASR rules targeting script-based and macro-based execution.
- 4) Monitor for anomalous parent-child process relationships in EDR.

HIGH**F-03 T1105 — Ingress Tool Transfer**

Command And Control

What happened Security control did not prevent this technique — it executed without being blocked.**Remediation**

- 1) Deploy DNS filtering to block known C2 domains and DGA-generated hostnames.
- 2) Enable TLS inspection at the perimeter for outbound HTTPS traffic.
- 3) Implement network detection for beaconing patterns (periodic, low-volume connections).
- 4) Restrict outbound connections to an approved destination allowlist.

HIGH**F-04 T1112 — Modify Registry**

Defense Evasion

What happened Security control did not prevent this technique — it executed without being blocked.**Remediation**

- 1) Audit and baseline security-relevant registry keys (Run keys, policy hives, ASR/Defender settings, LSA).
- 2) Enable registry-modification auditing (Sysmon Event IDs 12/13/14) and forward to SIEM.
- 3) Restrict write access to sensitive hives; alert on changes to Defender/EDR and policy keys.
- 4) Enable tamper protection so security-product registry settings cannot be silently altered.

HIGH**F-05 T1218.011 — System Binary Proxy Execution: Rundll32**

Defense Evasion

What happened Security control did not prevent this technique — it executed without being blocked.**Remediation**

- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
- 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
- 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
- 4) Monitor for security service termination and log clearing events.

HIGH**F-06 T1547.001 — Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder**

Persistence

What happened Security control did not prevent this technique — it executed without being blocked.**Remediation**

- 1) Monitor autorun locations: Run keys, scheduled tasks, services, WMI subscriptions.
- 2) Enable Autoruns/Sysmon monitoring with alerts on new persistence mechanisms.
- 3) Implement application allowlisting (WDAC or AppLocker) on all endpoints.
- 4) Audit and remediate all startup entries against a known-good baseline.

CRITICAL**F-07 T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control**

Privilege Escalation

What happened Security control did not prevent this technique — it executed without being blocked.**Remediation**

- 1) Enforce least-privilege; audit and remove unnecessary local admin memberships.
- 2) Enable UAC at highest level; require elevation approval for all admin actions.
- 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue.
- 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).

HIGH**F-08 T1569.002 — T1569.002 - Test 7: Modifying ACL of Service Control Manager via SDET**

Execution

What happened Security control did not prevent this technique — it executed without being blocked.**Remediation**

- 1) Audit service DACLS (sc.exe sdshow / SCM) and remove WRITE_DAC, WRITE_OWNER, and CHANGE_CONFIG from non-admin principals.
- 2) Harden SCM permissions; restrict who may create, reconfigure, or change the security descriptor of services.
- 3) Review SeTakeOwnershipPrivilege / SeRestorePrivilege assignments — restrict to trusted administrators.
- 4) Monitor service DACL changes and service-config modifications (Event IDs 7045, 4697) and alert on non-admin actors.

6. Detailed Technique Results

Results are grouped by ATT&CK technique. Each technique shows how many of its tests executed, were blocked, or errored. ERROR tests (a BAS execution problem, not a security outcome) are summarised in the counts and excluded from scoring.

T1003.001 — OS Credential Dumping: LSASS Memory

Credential Access · Critical · Active Adversary Behavioral Test

13 test(s): 0 executed · 6 blocked · 6 errored · 1 skipped

Threat impact	OS Credential Dumping: LSASS Memory (T1003.001): Attacker obtains valid credentials enabling lateral movement, privilege escalation, and persistent access. In BFSI environments this risks access to core banking, SWIFT, or payment switch systems.
Remediation	Control validated: OS Credential Dumping: LSASS Memory (T1003.001) was blocked. Maintain current security posture and continue monitoring.
PASS	Security control blocked the technique (access denied): fork/exec C:\Windows\system32\cmd.exe: Access is denied.
PASS	Security control blocked the technique (access denied): fork/exec C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe: Access is denied.
PASS	Security control blocked the technique (access denied): fork/exec C:\Windows\system32\cmd.exe: Access is denied.
PASS	Security control blocked the technique (access denied): fork/exec C:\Windows\system32\cmd.exe: Access is denied.
PASS	Security control blocked the technique (access denied): fork/exec C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe: Access is denied.
PASS	Security control blocked the technique (policy block): At line:1 char:1

T1003.002 — OS Credential Dumping: Security Account Manager

Credential Access · Critical · Active Adversary Behavioral Test

8 test(s): 0 executed · 3 blocked · 4 errored · 1 skipped

Threat impact	OS Credential Dumping: Security Account Manager (T1003.002): Attacker obtains valid credentials enabling lateral movement, privilege escalation, and persistent access. In BFSI environments this risks access to core banking, SWIFT, or payment switch systems.
Remediation	Control validated: OS Credential Dumping: Security Account Manager (T1003.002) was blocked. Maintain current security posture and continue monitoring.
PASS	Security control blocked the technique (access denied): Access is denied.
PASS	Security control blocked the technique (access denied): fork/exec C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe: Access is denied.
PASS	Security control blocked the technique (policy block): iex : At line:1 char:1

T1055.001 — Process Injection: Dynamic-link Library Injection

Privilege Escalation · Critical · Active Adversary Behavioral Test

2 test(s): 0 executed · 1 blocked · 0 errored · 1 skipped

Threat impact	Process Injection: Dynamic-link Library Injection (T1055.001): Attacker elevates to SYSTEM or Domain Admin enabling unrestricted access to all local and domain resources including AD, GPO, and privileged service accounts.
Remediation	1) Enforce least-privilege; audit and remove unnecessary local admin memberships. 2) Enable UAC at highest level; require elevation approval for all admin actions. 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue. 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).
PASS	Security control blocked the technique (policy block): iex : At line:1 char:1

T1059.001 — Command and Scripting Interpreter: PowerShell

Execution · High · Active Adversary Behavioral Test

21 test(s): 2 executed · 13 blocked · 4 errored · 2 skipped

Threat impact	Command and Scripting Interpreter: PowerShell (T1059.001): Attacker executes arbitrary code on the endpoint enabling full control of the compromised system and all data accessible from it.
Remediation	1) Enable PowerShell Script Block Logging and Constrained Language Mode. 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution. 3) Enable ASR rules targeting script-based and macro-based execution. 4) Monitor for anomalous parent-child process relationships in EDR.

PASS	Security control blocked the technique (access denied): fork/exec C:\Windows\system32\cmd.exe: Access is denied.
PASS	Security control blocked the technique (antivirus): Invoke-BloodHound : The term 'Invoke-BloodHound' is not recognized as the name of a cmdlet, function, script file, or
PASS	Security control blocked the technique (access denied): fork/exec C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe: Access is denied.
PASS	Security control blocked the technique (access denied): fork/exec C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe: Access is denied.
PASS	Security control blocked the technique (access denied): fork/exec C:\Windows\system32\cmd.exe: Access is denied.
PASS	Security control blocked the technique (access denied): fork/exec C:\Windows\system32\cmd.exe: Access is denied.
PASS	Security control blocked the technique (access denied): fork/exec C:\Windows\system32\cmd.exe: Access is denied.
PASS	Security control blocked the technique (access denied): fork/exec C:\Windows\system32\cmd.exe: Access is denied.
PASS	Security control blocked the technique (access denied): fork/exec C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe: Access is denied.
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed Evidence: Stream Data Executed
PASS	Security control blocked the technique (access denied): Source Destination IPV4Address IPV6Address Bytes Time(ms)
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed Evidence: Hello, from PowerShell! #< CLIXML <Objs Version="1.1.0.1" xmlns="http://schemas.microsoft.com/powershell/2004/04"><Obj S="progress" RefId="0"><TN RefId="0"><T>System.Management.Automation.PSCustomObject</T><T>System.Object</T></TN><MS><I64 ...
PASS	Security control blocked the technique (access denied): fork/exec C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe: Access is denied.
PASS	Security control blocked the technique (access denied): fork/exec C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe: Access is denied.
PASS	Security control blocked the technique (access denied): fork/exec C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe: Access is denied.

T1059.003 — Command and Scripting Interpreter: Windows Command Shell

Execution · High · Active Adversary Behavioral Test

6 test(s): 1 executed · 0 blocked · 2 errored · 3 skipped

Threat impact	Command and Scripting Interpreter: Windows Command Shell (T1059.003): Attacker executes arbitrary code on the endpoint enabling full control of the compromised system and all data accessible from it.
Remediation	1) Enable PowerShell Script Block Logging and Constrained Language Mode. 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution. 3) Enable ASR rules targeting script-based and macro-based execution. 4) Monitor for anomalous parent-child process relationships in EDR.
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed Evidence: Hello, from CMD!

T1082 — System Information Discovery

Discovery · Medium · Active Adversary Behavioral Test

28 test(s): 10 executed · 9 blocked · 6 errored · 3 skipped

Threat impact	System Information Discovery (T1082): Technique executed against the endpoint. Review result and apply recommended controls.
Remediation	1) Review and harden controls for System Information Discovery (T1082) based on vendor hardening guidance. 2) Enable relevant audit logging and forward to SIEM for correlation. 3) Apply CIS Benchmark or vendor-recommended hardening for the affected component. 4) Schedule follow-up simulation to verify remediation effectiveness.
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed Evidence: Host Name: WIN-MHQITQ23L3Q OS Name: Microsoft Windows Server 2019 Standard Evaluation OS Version: 10.0.17763 N/A Build 17763 OS Manufacturer: Microsoft Corporation OS Configuration: Standalone Server OS Build Type:

FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed Evidence: WIN-MHQITQ23L3Q
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed Evidence: HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Cryptography MachineGuid REG_SZ efce202b-f063-46fd-81e0-c22ca5347a60
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed Evidence: ALLUSERSPROFILE=C:\ProgramData APPDATA=C:\Windows\system32\config\systemprofile\AppData\Roaming BAS_PAYLOAD_DIR=C:\Windows\TEMP\bbs-18b7f39ddd2a441b-1623974456 BAS_RUN_MODE=posture CommonProgramFiles=C:\Program Files\Common Files CommonProg...
PASS	Security control blocked the technique (policy block): iex : At line:1 char:1
PASS	Security control blocked the technique (policy block): iex : At line:1 char:1
PASS	Security control blocked the technique (policy block): iex : At line:1 char:1
PASS	Security control blocked the technique (policy block): iex : At line:1 char:1
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PASS	Security control blocked the technique (policy block): iex : At line:1 char:1
PASS	Security control blocked the technique (policy block): iex : At line:1 char:1
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed Evidence: Name Intel(R) Core(TM) i7-7700 CPU @ 3.60GHz
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed Evidence: HKEY_LOCAL_MACHINE\HARDWARE\DESCRIPTION\System SystemBiosVersion REG_MULTI_SZ INTEL - 6040000\0VMW71.00V.7581552.B64.1801142334\0VMware, Inc. - 10000
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed Evidence: Microsoft Windows [Version 10.0.17763.3650]
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed Evidence: Get-Culture Format-List Out-File -FilePath C:\Windows\TEMP\la.txt
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed Evidence: gdr -PSProvider 'FileSystem'

T1087.001 — Account Discovery: Local Account

Discovery · Medium · Active Adversary Behavioral Test

4 test(s): 0 executed · 1 blocked · 3 errored · 0 skipped

Threat impact	Account Discovery: Local Account (T1087.001): Technique executed against the endpoint. Review result and apply recommended controls.
Remediation	1) Review and harden controls for Account Discovery: Local Account (T1087.001) based on vendor hardening guidance. 2) Enable relevant audit logging and forward to SIEM for correlation. 3) Apply CIS Benchmark or vendor-recommended hardening for the affected component. 4) Schedule follow-up simulation to verify remediation effectiveness.
PASS	Security control blocked the technique (antivirus): User accounts for \\\

Discovery · Medium · Active Adversary Behavioral Test

22 test(s): 1 executed · 2 blocked · 11 errored · 8 skipped

Threat impact	Account Discovery: Domain Account (T1087.002): Technique executed against the endpoint. Review result and apply recommended controls.
Remediation	1) Review and harden controls for Account Discovery: Domain Account (T1087.002) based on vendor hardening guidance. 2) Enable relevant audit logging and forward to SIEM for correlation. 3) Apply CIS Benchmark or vendor-recommended hardening for the affected component. 4) Schedule follow-up simulation to verify remediation effectiveness.
PASS	Security control blocked the technique (policy block): IEX : At line:1 char:1
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed
PASS	Security control blocked the technique (policy block): iex : At line:1 char:1

T1105 — Ingress Tool Transfer

Command And Control · High · Active Adversary Behavioral Test

29 test(s): 8 executed · 5 blocked · 11 errored · 5 skipped

Threat impact	Ingress Tool Transfer (T1105): Attacker establishes a C2 channel enabling persistent remote control, tool staging, and coordination of multi-stage attacks from outside the perimeter.
Remediation	1) Deploy DNS filtering to block known C2 domains and DGA-generated hostnames. 2) Enable TLS inspection at the perimeter for outbound HTTPS traffic. 3) Implement network detection for beaconing patterns (periodic, low-volume connections). 4) Restrict outbound connections to an approved destination allowlist.
PASS	Security control blocked the technique (access denied): fork/exec C:\Windows\system32\cmd.exe: Access is denied.
PASS	Security control blocked the technique (access denied): fork/exec C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe: Access is denied.
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed Evidence: BITSADMIN version 3.0 BITS administration utility. (C) Copyright Microsoft Corp. Transfer complete.
PASS	Security control blocked the technique (access denied): fork/exec C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe: Access is denied.
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed Evidence: 1 file(s) copied.
PASS	Security control blocked the technique (access denied): fork/exec C:\Windows\system32\cmd.exe: Access is denied.
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed Evidence: [WIN-MHQITQ23L3Q] > Finger: connect::Connection refused
PASS	Security control blocked the technique (access denied): fork/exec C:\Windows\system32\cmd.exe: Access is denied.
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed

T1112 — Modify Registry

Defense Evasion · High · Active Adversary Behavioral Test

90 test(s): 65 executed · 0 blocked · 25 errored · 0 skipped

Threat impact	Modify Registry (T1112): Attacker disables or bypasses security controls allowing subsequent techniques to execute undetected, defeating SIEM/SOAR alerting and EDR coverage.
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- 2) Enable registry-modification auditing (Sysmon Event IDs 12/13/14) and forward to SIEM.
- 3) Restrict write access to sensitive hives; alert on changes to Defender/EDR and policy keys.
- 4) Enable tamper protection so security-product registry settings cannot be silently altered.

FAIL

Security control did not prevent this technique — it executed without being blocked.

Detection: No detection telemetry observed

Evidence: The operation completed successfully.

FAIL

Security control did not prevent this technique — it executed without being blocked.

Detection: No detection telemetry observed

Evidence: The operation completed successfully.

FAIL

Security control did not prevent this technique — it executed without being blocked.

Detection: No detection telemetry observed

Evidence: The operation completed successfully.

FAIL

Security control did not prevent this technique — it executed without being blocked.

Detection: No detection telemetry observed

FAIL

Security control did not prevent this technique — it executed without being blocked.

Detection: No detection telemetry observed

Evidence: Hive: HKEY_CURRENT_USER\SOFTWARE\Microsoft\Windows\CurrentVersion\Internet Settings\ZoneMap\Domains\bad-domain.com Name Property ---- bad-subdomain https : 2 PSPath : Microsoft.PowerShell.Core\Registry::HKEY_CURRENT_USER\SOFTWARE\Microsoft\Windows\CurrentVersion\Internet Settings\ZoneMap\Domains\bad-domain.com Name Property ---- bad-subdomain https : 2 PSPath : Microsoft.PowerShell.Core\Registry::HKEY_CURRENT_USER\SOFTWARE\Microsoft\Windows\CurrentVersion\Internet Settings\ZoneMap\Domains\bad-domain.com Name Property ---- bad-subdomain https : 2 PSPath :

FAIL

Security control did not prevent this technique — it executed without being blocked.

Detection: No detection telemetry observed

Evidence: T1112 : <script> PSPath :

Microsoft.PowerShell.Core\Registry::HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Internet Settings PSParentPath : Microsoft.PowerShell.Core\Registry::HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Internet Settings PSParentPath : Microsoft.PowerShell.Core\Registry::HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Internet Settings PSParentPath :

FAIL

Security control did not prevent this technique — it executed without being blocked.

Detection: No detection telemetry observed

FAIL

Security control did not prevent this technique — it executed without being blocked.

Detection: No detection telemetry observed

Evidence: The operation completed successfully.

FAIL

Security control did not prevent this technique — it executed without being blocked.

Detection: No detection telemetry observed

Evidence: LocalAccountTokenFilterPolicy : 1 PSPath : Microsoft.PowerShell.Core\Registry::HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System PSParentPath : Microsoft.PowerShell.Core\Registry::HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System PSParentPath : Microsoft.PowerShell.Core\Registry::HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System PSParentPath :

FAIL

Security control did not prevent this technique — it executed without being blocked.

Detection: No detection telemetry observed

Evidence: The operation completed successfully.

FAIL

Security control did not prevent this technique — it executed without being blocked.

Detection: No detection telemetry observed

Evidence: The operation completed successfully.

FAIL

Security control did not prevent this technique — it executed without being blocked.

Detection: No detection telemetry observed

Evidence: The operation completed successfully.

FAIL

Security control did not prevent this technique — it executed without being blocked.

Detection: No detection telemetry observed

Evidence: The operation completed successfully.

FAIL

Security control did not prevent this technique — it executed without being blocked.

Detection: No detection telemetry observed

Evidence: The operation completed successfully.

FAIL

Security control did not prevent this technique — it executed without being blocked.

Detection: No detection telemetry observed

Evidence: The operation completed successfully.

FAIL

Security control did not prevent this technique — it executed without being blocked.

Detection: No detection telemetry observed

Evidence: The operation completed successfully.

FAIL

Security control did not prevent this technique — it executed without being blocked.

Detection: No detection telemetry observed

Evidence: The operation completed successfully.

[illegible]

[illegible]

FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed Evidence: The operation completed successfully.
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed Evidence: The operation completed successfully.
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed Evidence: The operation completed successfully.
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FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed Evidence: The operation completed successfully.
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed Evidence: The operation completed successfully.

T1134.001 — Access Token Manipulation: Token Impersonation/Theft

Privilege Escalation · Critical · Active Adversary Behavioral Test

5 test(s): 0 executed · 2 blocked · 1 errored · 2 skipped

Threat impact	Access Token Manipulation: Token Impersonation/Theft (T1134.001): Attacker elevates to SYSTEM or Domain Admin enabling unrestricted access to all local and domain resources including AD, GPO, and privileged service accounts.
Remediation	Control validated: Access Token Manipulation: Token Impersonation/Theft (T1134.001) was blocked. Maintain current security posture and continue monitoring.
PASS	Security control blocked the technique (policy block): IEX : At line:1 char:1
PASS	Security control blocked the technique (policy block): IEX : At line:1 char:1

T1218.011 — System Binary Proxy Execution: Rundll32

Defense Evasion · High · Active Adversary Behavioral Test

16 test(s): 4 executed · 3 blocked · 1 errored · 8 skipped

Threat impact	System Binary Proxy Execution: Rundll32 (T1218.011): Attacker disables or bypasses security controls allowing subsequent techniques to execute undetected, defeating SIEM/SOAR alerting and EDR coverage.
Remediation	1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement. 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI. 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM. 4) Monitor for security service termination and log clearing events.
PASS	Security control blocked the technique (access denied): fork/exec C:\Windows\system32\cmd.exe: Access is denied.
PASS	Security control blocked the technique (access denied): fork/exec C:\Windows\system32\cmd.exe: Access is denied.
PASS	Security control blocked the technique (access denied): fork/exec C:\Windows\system32\cmd.exe: Access is denied.
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed Evidence: 1 file(s) copied.
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed

Persistence · High · Active Adversary Behavioral Test

20 test(s): 9 executed · 0 blocked · 6 errored · 5 skipped

Threat impact	Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder (T1547.001): Attacker maintains long-term access surviving reboots and credential rotations, enabling sustained espionage or pre-positioned ransomware in RBI-regulated environments.
Remediation	1) Monitor autorun locations: Run keys, scheduled tasks, services, WMI subscriptions. 2) Enable Autoruns/Sysmon monitoring with alerts on new persistence mechanisms. 3) Implement application allowlisting (WDAC or AppLocker) on all endpoints. 4) Audit and remediate all startup entries against a known-good baseline.
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed Evidence: The operation completed successfully.
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed Evidence: The operation completed successfully.
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed Evidence: Directory: C:\Windows\TEMP Mode LastWriteTime Length Name ---- - 6/10/2026 11:56 PM atomictest
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed

T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control

Privilege Escalation · Critical · Active Adversary Behavioral Test

27 test(s): 10 executed · 5 blocked · 11 errored · 1 skipped

Threat impact	Abuse Elevation Control Mechanism: Bypass User Account Control (T1548.002): Attacker elevates to SYSTEM or Domain Admin enabling unrestricted access to all local and domain resources including AD, GPO, and privileged service accounts.
Remediation	1) Enforce least-privilege; audit and remove unnecessary local admin memberships. 2) Enable UAC at highest level; require elevation approval for all admin actions. 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue. 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed Evidence: The operation completed successfully.
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed Evidence: Hive: HKEY_CURRENT_USER\software\classes\mscfile\shell\open Name Property ---- - command
PASS	Security control blocked the technique (access denied): fork/exec C:\Windows\system32\cmd.exe: Access is denied.
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed Evidence: Hive: HKEY_CURRENT_USER\software\classes\ms-settings\shell\open Name Property ---- - command DelegateExecute : PSPath : Microsoft.PowerShell.Core\Registry::HKEY_CURRENT_USER\software\classes\ms-settings\shell\open\command PSParentPat...
FAIL	Security control did not prevent this technique — it executed without being blocked. Detection: No detection telemetry observed Evidence: Hive: HKEY_CURRENT_USER\software\classes\ms-settings\shell\open Name Property ---- - command

Microsoft.PowerShell.Core\Registry::HKEY_CURRENT_USER\software\classes\ms-settings\shell\open\command
PSParentPat...

FAIL

Security control did not prevent this technique — it executed without being blocked.

Detection: No detection telemetry observed

Evidence: The operation completed successfully.

PASS

Security control blocked the technique (policy block): iex : At line:1 char:1

PASS

Security control blocked the technique (policy block): iex : At line:1 char:1

PASS

Security control blocked the technique (policy block): iex : At line:1 char:1

PASS

Security control blocked the technique (policy block): iex : At line:1 char:1

FAIL

Security control did not prevent this technique — it executed without being blocked.

Detection: No detection telemetry observed

FAIL

Security control did not prevent this technique — it executed without being blocked.

Detection: No detection telemetry observed

FAIL

Security control did not prevent this technique — it executed without being blocked.

Detection: No detection telemetry observed

FAIL

Security control did not prevent this technique — it executed without being blocked.

Detection: No detection telemetry observed

Evidence: The operation completed successfully.

FAIL

Security control did not prevent this technique — it executed without being blocked.

Detection: No detection telemetry observed

Evidence: The operation completed successfully.

T1569.002 — T1569.002 - Test 1: Execute a Command as a Service

Execution · High · Active Adversary Behavioral Test

7 test(s): 1 executed · 0 blocked · 3 errored · 3 skipped

Threat impact

T1569.002 - Test 7: Modifying ACL of Service Control Manager via SDET (T1569.002): Attacker executes arbitrary code on the endpoint enabling full control of the compromised system and all data accessible from it.

Remediation

- 1) Audit service DACLS (sc.exe sdshow / SCM) and remove WRITE_DAC, WRITE_OWNER, and CHANGE_CONFIG from non-admin principals.
- 2) Harden SCM permissions; restrict who may create, reconfigure, or change the security descriptor of services.
- 3) Review SeTakeOwnershipPrivilege / SeRestorePrivilege assignments — restrict to trusted administrators.
- 4) Monitor service DACL changes and service-config modifications (Event IDs 7045, 4697) and alert on non-admin actors.

FAIL

Security control did not prevent this technique — it executed without being blocked.

Detection: No detection telemetry observed

Evidence: [SC] SetServiceObjectSecurity SUCCESS

7. Changes & Cleanup Verification

The agent reverted 1 endpoint change(s) made during the run. Cleanup status: COMPLETED — each captured change was rolled back from the pre-run snapshot.

- registry removed: HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run\NetWire

Scope note: residual-artifact tracking covers changes captured by the pre-run snapshot (registry keys and files the engine instruments). Out-of-band changes a technique may make outside that scope are not tracked here — verify manually for high-impact techniques run in production.

8. Prioritised Recommendations

- 1 Overall prevention score is critically low. Prioritise endpoint hardening and EDR deployment across all assets.
- 2 Critical tactic 'privilege-escalation' has 10 failing technique(s). Address immediately — this tactic has the highest attacker value in BFSI environments.

- 3 **Kill-chain amplifier** is 1.8× — an attacker can traverse multiple consecutive kill-chain phases unimpeded. Implement network segmentation and lateral movement controls.
- 4 Exposure score exceeds 60. Deploy a 24×7 SOC with SIEM correlation rules aligned to detected failure patterns.

9. Appendix — Metric Definitions

PASS	A security control prevented or blocked the simulated technique. More passes is better.
FAIL	A prevention control did not stop the technique from executing — a finding requiring attention. This means execution was permitted; it does NOT independently verify the attacker's end objective was achieved (e.g. that credentials were actually exfiltrated), which is technique-specific. Check the Detection line for whether the execution was also detected.
Detection	Whether the executed technique was seen by locally-observable telemetry: Detected (a Microsoft Defender alert fired), Logged only (Sysmon/Security telemetry exists but no alert), or Undetected (no telemetry). Third-party EDR detection is not locally observable and is never asserted.
ERROR	The test could not execute correctly (malformed content, timeout, missing prerequisite, scheduler contention). A BAS execution problem, not a security outcome — excluded from scoring.
SKIPPED	The technique was deliberately not run (e.g. external payload not shipped) and was excluded from scoring.
Policy Configuration Check	A passive audit that inspects a security setting (registry key, policy, service state) without running an attack — it confirms whether a control is correctly configured.
Active Adversary Behavioral Test	An ART or Caldera test that actually executes the technique on the endpoint — it confirms whether deployed controls stop a live attack, not just whether they are configured.
Prevention	Severity-weighted percentage of techniques that were prevented. Higher is better.
Exposure	Tactic-weighted failure rate, amplified by consecutive kill-chain failures. Lower is better.
Tactic Breadth	Breadth of THIS run: how many of the 14 MITRE ATT&CK Enterprise tactics it exercised. Not a measure of overall MITRE ATT&CK technique coverage — running more Atomics in a tactic does not increase ATT&CK coverage.
Defense Rate	Share of tested tactics in which every technique was blocked (zero failures).
Kill-chain amplifier	Multiplier (1.0–2.5×) reflecting consecutive unprevented kill-chain phases.
Risk classification	Overall posture band derived from the amplified exposure score.